LaTeXand Markdown

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What is a Markup Language?

- HTML
- Markdown
- LaTeX
- XML

LaTeX

```
\begin{itemize}
    \item First item
    \item Second item
    \item Third item
\end{itemize}
```

Used to help format text in a universal structure.
Easy to share as everything is nearly plain text.

Markdown

Markdown was invented to be a simple format that is easily parseable by any application. It is used in software such as:

- Google Docs
- Obsidian
- GitHub
- Stack Exchange

Markdown

```
## Quick start
1. Some list first column
2. something ‘important‘
‘‘‘python
some code
‘‘‘
```

Pros and Cons

Pros:

- Easy to read documents
- Well organized data
- Somewhat readable even without using a viewer
- Open Source

Cons:

- More software requirements
- Each standard has different limitations
- Can be complex to learn for the first time

The Power of LaTeX

- Significant Math Power
- Used in technical presentations
- Turing Complete?

3.3 Quantum Gates

While states can be defined as vectors, a gate may be defined as a matrix. Simply, a two by two matrix is used to define a single qubit gate.

$$U = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

This is particularly useful because we can see how our states ($|0\rangle$ and $|1\rangle$) are just basis vectors that when multiplied by our gate will return us a single column of that gate. We can now define many of our gates as matrices.

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, T = \begin{pmatrix} 1 & 0 \\ 0 & e^{\frac{i\pi}{4}} \end{pmatrix}$$

3.4 Outer Products

In addition to the inner products defined in the previous section, there are also outer products that can be defined as $|\phi\rangle \langle\psi|$. This results in the following:

$$|\phi\rangle \langle\psi| = \begin{pmatrix} \alpha\gamma^* & \alpha\delta^* \\ \beta\gamma^* & \beta\delta^* \end{pmatrix}$$

Additionally, we can show that based on our assertion about outer products, it must follow that for some $|\psi\rangle = \alpha |a\rangle + \beta |b\rangle$, $|a\rangle \langle a| + |b\rangle \langle b| = I$.

4 Multiple Quantum Bits

4.1 Entanglion

It so seems that there are multiple quantum computing based board games.

4.2 States and Measurement

We must now define our final form of product, the tensor product. We let $|0\rangle \otimes |0\rangle = |0\rangle |0\rangle = |00\rangle$. With two qubits, we are now given four options for our tensors. $|00\rangle, |01\rangle, |10\rangle, |11\rangle$. We can now define a superposition of two qubits in the z-basis to be $c_0 |00\rangle + c_1 |01\rangle + c_2 |10\rangle + c_3 |11\rangle$ where the probability of measuring any one of these states is equal to the norm-square of its respective c coefficient. For a more explicit definition of the tensor product, we can view it as:

$$\begin{pmatrix} \alpha \\ \beta \end{pmatrix} \otimes \begin{pmatrix} \gamma \\ \delta \end{pmatrix} = \begin{pmatrix} \alpha \begin{pmatrix} \gamma \\ \delta \end{pmatrix} \\ \beta \begin{pmatrix} \gamma \\ \delta \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \alpha\gamma \\ \alpha\delta \\ \beta\gamma \\ \beta\delta \end{pmatrix}$$

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Conclusion

- Markup Languages are very powerful for rendering text in comprehensible formats.
- These formats can be used for many tasks, from taking notes, to hosting websites.
- Tools like Pandoc can be used to transform between formats.
- Organizations like arXiv support multiple formats depending on user preference.